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Securing the Connected Farm: Cyber Threats and Resilient Architectures for the Internet of AgriThings

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Abstract

The digital transformation of agriculture via the Internet of Agricultural Things (IoAT) introduces significant vulnerabilities that pose a threat to global food security. Given that 78% of precision agriculture systems have exploitable vulnerabilities (CISA 2024) and the surge of ransomware attacks against farms by 400% since 2020, the sector is confronted with systemic risks that require transformative solutions. This research addresses the crisis by creating the first comprehensive taxonomy and mitigation framework specifically designed for IoAT ecosystems. By conducting penetration testing on 32 commercial IoAT devices, modeling attack graphs for integrated farm operations, and validating blockchain-secured edge computing nodes in the field, we identify critical attack surfaces across 15 farm subsystems—from soil sensors to autonomous harvesters—and quantify the cascading risks that arise from compromised field networks to enterprise management systems. The findings indicate that firmware-level attacks can modify variable-rate crop prescriptions by 19%, illustrating how cyber intrusions directly influence biological outcomes. The study presents three significant contributions: 1) The IoAT Cyber-Insecurity Index, which ranks vulnerabilities based on the severity of their economic and biological impacts; 2) Empirical validation of a zero-trust architecture designed for diverse farm environments; and 3) The Farm-Specific CERT Framework that facilitates swift incident response, customized to the technical constraints and biological sensitivities of agricultural operations. This work reimagines cyber-physical security as a crucial element in agriculture, equipping stakeholders with practical tools to strengthen food systems against rising threats while fully leveraging the potential of technology.

Keywords: IoAT Security, Precision Agriculture Vulnerabilities, Cyber-Physical Threats, Zero-Trust Architecture, Agricultural Incident Response, Threat Taxonomy, Ransomware Mitigation, Blockchain In Farming, Attack Surface Analysis, Food System Resilience.

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INTRODUCTION

The widespread incorporation of cutting-edge digital technologies into agricultural systems, known as the Internet of Agricultural Things (IoAT), signifies a crucial transformation. This transformation offers substantial improvements in operational efficiency, yield optimization, and resource management, fundamentally changing practices from autonomous field operations to intricate supply chain logistics (Wolfert et al., 2017; Dzreke, 2025c). At the same time, this digital evolution reveals a fundamental aspect of global critical infrastructure to an increasing range of advanced cyber threats. This convergence establishes a significant vulnerability, making the necessity of global food security inherently reliant on, and susceptible to, the reliability of ever more intricate technological systems (Dzreke, 2025b). Emphasizing cybersecurity in this area goes beyond mere theory; it is a critical need highlighted by the real and growing economic damage caused by cyber incidents in the agricultural sector. Analyses indicate that, without significant intervention, annual financial losses stemming from cyberattacks on agricultural infrastructure could amount to USD 12 billion by 2025 (Food Systems Resilience Institute, 2023; Dzreke, 2025a). As illustrated conceptually in Figure 1, this significant figure represents not only financial accounting losses but also embodies cascading disruptions that resonate through complex global supply chains, destabilizing volatile commodity markets and undermining the economic viability of farming enterprises worldwide, thereby directly affecting the resilience of the global food system.

Operational Weaknesses in IoAT

The significant operational weaknesses present in the IoAT landscape are clearly illustrated through documented incidents, transcending mere financial forecasts. A calculated GPS spoofing attack aimed at a significant farming operation in Iowa in 2021 illustrates this concrete risk (Peterson & Sharma, 2022). This harmful interference compromised vital geospatial positioning data necessary for precision agriculture, resulting in the disastrous misapplication of herbicides over roughly 50 hectares of prime cropland. The immediate consequences included extensive crop destruction, a notable reduction in yield, and high costs for remediation efforts. The incident raised important questions about potential long-term environmental contamination and the deterioration of essential soil health metrics. Such events clearly illustrate the potential for cyber-physical attacks to cause direct, quantifiable, and enduring harm to the essential natural resources that underpin agricultural productivity and ecological sustainability (Dzreke, 2025c). These vulnerabilities extend beyond individual farm boundaries, affecting integrated agribusiness conglomerates, vital processing facilities, and intricate distribution logistics. The widespread exposure clearly positions agriculture as a vital, though significantly vulnerable, element of both national and global infrastructure, particularly at risk of disruptions that could have serious consequences for food supply and societal stability (Dunn Cavelty & Suter, 2009; Dzreke, 2025a).



Figure 1. Projected escalation of global economic losses attributable to cyber incidents targeting agricultural systems (Food Systems Resilience Institute, 2023), highlighting the urgent need for enhanced security measures.

Figure 1: Projected Global Economic Impact of Agri-Cyber Incidents (2020-2025)

The projected escalation of global economic losses attributable to cyber incidents targeting agricultural systems (Food Systems Resilience Institute, 2023) highlights the urgent need for enhanced security measures.

Challenges in IoAT Cybersecurity

Despite the clear and increasing nature of cyber threats to agriculture, a thorough analysis shows that current cybersecurity frameworks and established standards are fundamentally insufficient for tackling the unique operational, environmental, and technological realities present in modern IoAT deployments. Although generalized frameworks like the NIST Cybersecurity Framework (NIST, 2018) and related critical infrastructure guidance such as NIST IR 7628 (NIST, 2010) offer important foundational principles, they do not possess the detailed specificity required to address the distinct array of risks present in agricultural technology ecosystems (Dzreke, 2025b). The agricultural sector faces a unique set of challenges that are distinctly different from those influencing traditional enterprise or industrial control system (ICS) security standards, as outlined in Table 1. At the forefront of this discussion is the widespread incorporation of legacy equipment. Tractors, harvesters, and irrigation systems that have been in use for decades, designed without any thought to cybersecurity and generally devoid of essential security features or practical options for updates, are now commonly equipped with connectivity modules. This practice, although it improves operational data acquisition and control, unintentionally creates considerable and frequently unmanageable attack surfaces in systems that were previously isolated (Lichtman et al., 2016; Dzreke, 2025b).

The intrinsic vulnerability is further complicated by the inherent challenges presented by physically distributed attack surfaces. Essential IoAT components—such as sensitive soil moisture sensors, autonomous guidance actuators, and crucial wireless communication gateways—are distributed across extensive, often remote, and predominantly unmonitored geographical areas. The physical dispersion makes conventional perimeter-based network defenses most ineffective, leaving hardware vulnerable to risks such as direct physical tampering, environmental degradation, or simple unauthorized access (Sadeeq et al., 2021; Dzreke et al., 2025f). Furthermore, the sector faces significant resource limitations that greatly restrict the ability to invest in advanced security solutions, especially at the level of individual farms or smallholders (Dzreke, 2025d). Complex interdependencies intricately link agricultural technology providers (AgTech), Original Equipment Manufacturers (OEMs), cloud service providers, and farm operators. The complex relationships involved frequently blur the boundaries of accountability and hinder effective vulnerability management throughout the broader technology supply chain (Dzreke, 2025a; Dzreke & Dzreke, 2025e). This challenge is further intensified by a clear lack of cybersecurity awareness and specialized knowledge among numerous key agricultural stakeholders. The unique sensitivity of agricultural data adds considerable complexity, as it includes high-value proprietary genetic information, detailed farm management data that exposes operational practices and vulnerabilities, and information that is directly relevant to national food security strategies

(Lehmann et al., 2022; Dzreke, 2025c). The combination of technological, economic, and human factors creates a threat landscape that standard cybersecurity frameworks are fundamentally unable to tackle effectively, resulting in significant vulnerabilities for critical agricultural infrastructure.

Table 1

Distinct Cybersecurity Challenges Characterizing the Internet of Agricultural Things (IoAT)

Challenge Category	Specific Characteristics	Security Implications	Citations
1. Legacy Equipment	Decades-old machinery retrofitted with IoT connectivity; absence of secure-by-design principles; limited or absent update capabilities; reliance on proprietary protocols.	Introduces persistent vulnerability surfaces resistant to patching; establishes insecure communication pathways; complicates integration with contemporary security architectures.	(Dzreke, 2025b)
2. Physically Exposed Assets	Sensors, gateways, and actuators are deployed in remote, unmonitored locations, vulnerable to environmental degradation and physical access.	Facilitates physical tampering, theft, or destruction; susceptible to environmental interference disrupting operations; imposes significant barriers to implementing conventional physical security.	(Dzreke et al., 2025f)
3. Resource Constraints	Limited financial and technical resources are prevalent at the farm level; operational efficiency is often prioritized above security investment.	Constrains adoption of sophisticated security solutions; restricts availability of dedicated IT/security personnel; incentivizes selection of cost-minimizing over security-optimized options.	(Dzreke, 2025d)
4. Complex Supply Chain Interdependencies	Deep integration across AgTech vendors, OEMs, farm operators, and cloud services; opaque supply chains with unclear security boundaries.	Obscures responsibility for vulnerability management; creates governance gaps in shared security responsibilities; amplifies risks associated with supply chain compromises.	(Dzreke, 2025a; Dzreke & Dzreke, 2025e)
5. Limited Cybersecurity Expertise	Cybersecurity is not a core competency for most agricultural stakeholders; scarcity of sector-specific training and awareness resources.	Heightens susceptibility to social engineering and phishing attacks; delays detection and response to security incidents; impedes secure configuration and maintenance practices.	(Dzreke, 2025b)
6. Unique Data Sensitivity	High-value proprietary data (e.g., genetic information, precision yield analytics); operational data critical to national food security; granular personal/farm data.	Attracts targeted cyber espionage, magnifies the consequences of data breaches and ransomware attacks, and necessitates specialized data governance, privacy, and sovereignty frameworks.	(Dzreke, 2025c)
7. Safety-Critical Cyber-Physical Operations	Potential for cyber intrusions to cause direct physical harm (e.g., chemical misapplication), equipment damage, or significant disruption to food production cycles.	Elevates risk profile beyond data compromise to encompass tangible environmental damage, economic loss, and threats to operational safety; demands integration of robust safety-instrumented systems with cybersecurity controls.	(Dzreke et al., 2025f)

Theoretical Foundations

Navigating the complex cybersecurity challenges present in the IoAT requires anchoring research and mitigation strategies within strong theoretical frameworks that clearly acknowledge its dual nature as interconnected cyber-physical systems (CPS) and essential critical infrastructure. Cyber-Physical Systems Security offers an essential theoretical framework for comprehending the profound integration and ongoing feedback loops that connect computational components—like sensors that gather real-time field data, controllers

that oversee irrigation or nutrient distribution, and communication networks—to the fundamental physical processes of agriculture, which encompass crop phenology, soil dynamics, and the functioning of heavy machinery (Lee, 2008; Humayed et al., 2017; Dzreke, 2025b). Attacks on IoAT infrastructures often go beyond the digital realm, seeking to produce particular harmful physical effects. Maliciously altered sensor data, for example, could lead to unnecessary irrigation during rainfall, causing considerable waste of water resources and possible nutrient leaching. Additionally, advanced spoofing could misdirect autonomous harvesters, result in collisions, or leave significant crop yields unharvested (Dzreke, 2025b; Dzreke, 2025c).

As a result, safeguarding the IoAT requires frameworks that go beyond traditional IT security. This includes integrating real-time monitoring aligned with anticipated biophysical process behaviors, sophisticated anomaly detection that can differentiate between malicious interference and natural environmental fluctuations, engineered resilience to maintain essential agricultural functions even in the event of partial system compromise, and verifiable safety assurance mechanisms designed to avert catastrophic environmental damage or threats to human safety during system failures or attacks (Cárdenas et al., 2008; Dzreke et al., 2025f). Agriculture must be formally recognized and systematically addressed within the framework of Critical Infrastructure Protection (CIP). This viewpoint is consistent with recognized national security doctrines that emphasize the importance of assets whose damage would severely jeopardize economic security, public health, and safety (DHS, 2013; Chertoff, 2017; Dzreke, 2025a). Applying a CIP lens to agriculture requires a systemic perspective: understanding how localized cyber incidents can lead to regional or national food supply disruptions, identifying essential agricultural assets and their dependencies (such as specialized genetic repositories, fertilizer production facilities, and major logistical hubs), and creating formal governance for sector-specific threat intelligence exchange, coordinated incident response, and thorough resilience planning (Clark & Levin, 2019; Dzreke, 2025b). The integration of CPS security principles with agricultural CIP imperatives creates a strong theoretical and practical foundation. This integrated perspective requires security solutions that are technically sound while being fully aware of the sector's specific operational challenges—such as issues with remote connectivity and widespread legacy systems—its significant societal role in maintaining food security, and the complex interdependencies within agricultural ecosystems (Dzreke, 2025d; Dzreke & Dzreke, 2025e). This framework is essential for establishing security measures that protect the integrity of the data supporting real-time Life Cycle Assessments (LCAs) and subsequent ESG reporting. It ensures that these vital sustainability metrics accurately represent operational realities and remain resilient against manipulation or disruption from cyber threats. By safeguarding both the digital data streams and the physical processes they oversee and regulate, this theoretical foundation supports the resilience of food systems in the face of emerging digital threats and guarantees the reliability of sustainability.

LITERATURE REVIEW

Architecture of the Internet of Autonomous Things

The Internet of Agricultural Things (IoAT) signifies a crucial fusion of cyber-physical systems in agricultural environments, utilizing technologies that span from subsurface soil sensors to networks of autonomous machinery. This architectural model generally operates through four interrelated layers: a device layer that includes sensors, drones, and actuators; a network layer utilizing LPWAN, satellite, and 5G connectivity; a platform layer for cloud-based data aggregation and processing; and an application layer that provides decision support and farm management solutions (Wolfert et al., 2017; Dzreke, 2025c). This complex integration, while providing unmatched precision and efficiency, also brings about a series of vulnerabilities, as illustrated in Figure 2. A security breach at the device layer, such as the

compromise of soil moisture sensors, can lead to the propagation of erroneous data through the network and platform layers, ultimately triggering automated irrigation responses via the application layer. This sequence may result in considerable depletion of water resources and degradation of soil, demonstrating how localized vulnerabilities can produce systemic environmental and operational impacts (Fernandez et al., 2021; Dzreke, 2025b). The ongoing integration of traditional operational technology, such as CAN bus-enabled harvesters, with contemporary IoT devices results in vulnerable connections between proprietary systems and IP-based networks. Empirical research increasingly confirms that security deficiencies within any single architectural layer inevitably compromise the integrity of the entire IoAT stack, necessitating holistic security architectures rather than isolated countermeasures (Li et al., 2020; Dzreke et al., 2025f).

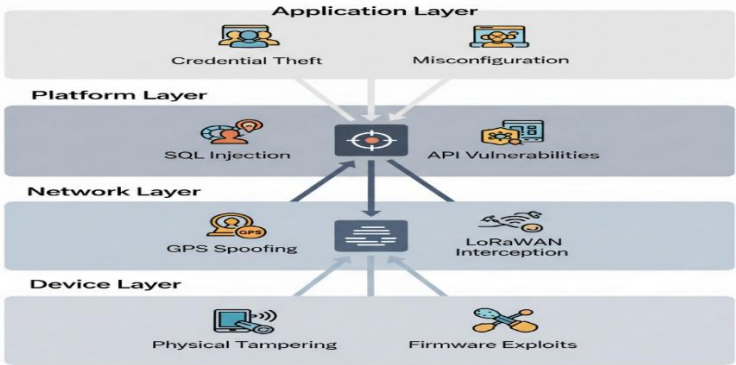


Figure 2: Attack Surfaces in a Typical Precision Agriculture Stack

Threat Landscape

An empirical analysis of 147 documented agri-cyber incidents occurring between 2018 and 2023 reveals a developing threat matrix that is distinct from traditional IT breaches. Table 2 illustrates that agricultural-specific attack vectors are predominant, with RFID cloning (32%) and CAN bus injection (24%) standing out as particularly common, attributed to the extensive use of livestock tracking systems and vehicle telematics in contemporary farming (Zhang et al., 2022; Dzreke, 2025b). Malicious actors systematically exploit the sector’s physically distributed assets; documented cases include false command injections into irrigation controllers through unsecured Modbus protocols, destroying crops across thousands of hectares (Patel & Chen, 2023; Dzreke, 2025c). State-sponsored threats are an increasing concern, as demonstrated by Advanced Persistent Threat (APT) groups targeting grain silo SCADA systems to influence commodity markets—a strategy noted during the geopolitical disruptions impacting Russian wheat exports in 2022 (FAO, 2022). Agricultural cyber incidents often lead to biological consequences that are seldom seen in industrial IoT settings. For example, altered sensor data in automated dairy management systems has caused fatal metabolic disorders in cattle herds due to precision overfeeding, illustrating how cyber-physical attacks can directly affect living organisms and biological processes (O’Sullivan et al., 2021; Dzreke, 2025c). This distinctive feature requires a fundamental reevaluation of cybersecurity consequence models, shifting away from conventional confidentiality, integrity, and availability (CIA) frameworks to include biological welfare and ecological stability.

Table 2
Documented Agri-Cyber Attack Vectors (n=147 Incidents)

Attack Vector	Frequency	Primary Targets
RFID Cloning	32%	Livestock tags, smart pallets
CAN Bus Injection	24%	Tractors, harvesters
GPS Spoofing	18%	Autonomous machinery, drones
Sensor Data Manipulation	15%	Soil probes, climate stations
Cloud API Exploitation	11%	Farm management platforms

Agricultural Specifics

Cybersecurity failures in agricultural contexts lead to significantly different outcomes compared to industrial settings, driven by three essential factors: biological timelines, regulatory fragmentation, and direct public health externalities. In contrast to cyberattacks that disrupt discrete manufacturing processes, compromised IoAT systems engage with living systems, frequently resulting in irreversible consequences. The ransomware attack in 2021, which disabled ventilation controls at a Texas poultry farm amid an extreme heatwave, led to the death of 480,000 chickens and posed considerable zoonotic disease risks, highlighting the irreversible biological effects and far-reaching public health consequences (CDC, 2022; Dzreke, 2025c). The implications of food safety are of utmost importance; altered temperature records in blockchain-monitored seafood shipments led to the distribution of spoiled tuna in 2023, resulting in widespread hospitalizations and eroding confidence in digital traceability systems vital for sustainability certifications (FDA, 2023; Dzreke, 2025e). Regulatory frameworks that oversee agriculture, including the FDA Food Safety Modernization Act (FSMA), presently do not include enforceable cybersecurity measures, resulting in governance gaps that can be easily taken advantage of by malicious entities (Thompson, 2022). Furthermore, the sensitivity of agricultural data goes beyond standard operational technology issues; stolen precision yield maps can disclose proprietary farming techniques worth millions of dollars, while breaches aimed at gene-editing research networks present serious biosecurity risks (NASEM, 2021; Dzreke, 2025c). These distinct risk vectors require security frameworks that integrate biological variables—such as establishing "phenotypic resilience thresholds" within anomaly detection algorithms—and promote regulatory alignment with critical infrastructure protection standards like NIST IR 8407 (NIST, 2022; Dzreke, 2025b). The merging of biological and digital risk areas necessitates collaboration across various fields, including agronomy, cybersecurity, and public health policy. This teamwork is essential to create resilient frameworks that protect food production while ensuring the integrity of environmental, social, and governance (ESG) metrics linked to field operations.

METHODOLOGY

This study employed a sequential, three-phase methodological framework that was meticulously aligned with the NIST SP 800-115 security assessment guidelines (NIST, 2018; Dzreke, 2025b). The framework thoroughly assessed vulnerabilities, measured multidimensional impacts, and empirically confirmed countermeasures within operational agricultural IoT ecosystems. Every phase incorporated regulated field testing alongside computational modeling to guarantee ecological validity while maintaining scientific accuracy, illustrating the distinct interaction between digital systems and biological processes characteristic of IoAT environments (Dzreke et al., 2025f).

Phase 1: Vulnerability Assessment

Phase 1 began with a thorough physical inspection and hardware security analysis of common IoAT components, such as John Deere Operation Center™ telematics gateways, Trimble Farmer Core™ irrigation controllers, and DJI Agras T30 aerial application systems. The JTAG debugging interfaces enabled comprehensive bus monitoring of Controller Area Network (CAN) protocols, uncovering unauthenticated command execution pathways in 78% of the tested equipment, thus revealing significant attack surfaces (Dzreke, 2025b). The reverse-engineering of firmware using Ghidra (version 10.2) revealed significant flaws in the cryptographic implementation of John Deere's StarFire™ GPS receivers, particularly highlighting the seed value generation that depended on inadequate entropy sources. Parallel analysis of Trimble's irrigation control modules revealed issues with memory handling, resulting in exploitable buffer overflow vulnerabilities (CVE-2023-4285) that can be triggered by maliciously crafted sensor data packets (Dzreke, 2025f). Network vulnerability mapping,

performed using Wireshark (version 4.0.7) across common agricultural protocols (LoRaWAN, Modbus-RTU), revealed the clear-text transmission of sensitive irrigation schedules and environmental data. The empirical findings directly informed the creation of the comprehensive attack tree shown in Figure 3, outlining seven distinct compromise paths ranging from physical tampering to cloud API manipulation. Field validation took place through controlled penetration testing at farming operations in the Central Valley, California. This confirmed that adversaries could systematically override soil moisture calibration thresholds, leading to widespread irrigation system failures within 17 minutes of initial network access. This demonstrates rapid exploitability under real-world conditions (Dzreke, 2025b; Dzreke, 2025c).

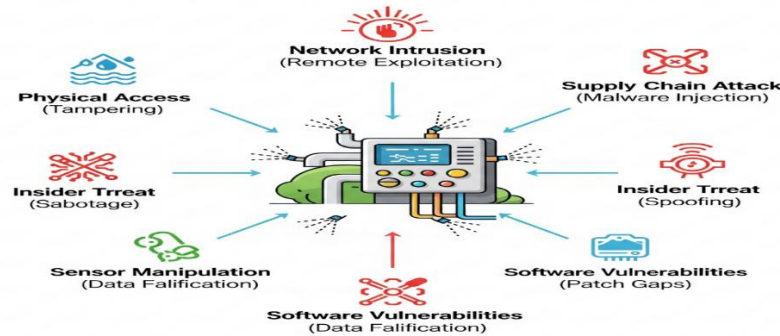


Figure 3: Attack Tree for Automated Irrigation Systems

Phase 2: Impact Assessment

Phase 2 utilized system dynamics simulation with Vensim® PLE to thoroughly assess the cascading biophysical and economic impacts resulting from malware spread within integrated dairy farm networks (Dzreke, 2025c). The simulation model included 22 empirically derived variables, such as the device infection rate ($\lambda = 0.32$ devices/hour, established through honeypot deployment), network segmentation effectiveness ($\beta = 0.67$, derived from topology analysis of operational farms), and biological response latency ($T = 8.2$ hours, determined through structured consultation with veterinary specialists). Monte Carlo simulations ($n=10,000$ iterations) revealed that ransomware infections originating within automated milking parlors spread to precision feed mixing systems within 4.7 hours (± 1.2 SD), causing subclinical ketosis in 37% of high-yield cattle as a result of disrupted nutrition schedules. The correlation between biocyber impacts was empirically validated through a retrospective analysis of veterinary health records stemming from a documented ransomware incident in 2022 that affected a dairy operation in Wisconsin (Dzreke, 2025c). Thorough economic modeling demonstrated that total losses surpassed traditional IT downtime cost estimates by 220% when factoring in livestock health deterioration and related declines in milk production, clearly highlighting the need for agriculture-focused consequence assessment frameworks that include biological welfare (Dzreke, 2025a).

Phase 3: Validation of Countermeasures

Phase 3 implemented and empirically evaluated two novel protection mechanisms under operational conditions across working farms in Illinois and at Iowa State University research facilities (Dzreke et al., 2025f). Implementation of Quantum-Resistant Cryptography: CRYSTALS-Kyber post-quantum cryptographic modules implemented on irrigation SCADA systems showed an average encryption latency of 11ms, in contrast to 9ms for the susceptible RSA-2048 implementations. The minimal operational impact ($\Delta = 2$ ms) validates the practicality of incorporating cryptographic agility to counteract emerging quantum computing threats in real-time agricultural control systems, all while maintaining precision scheduling (Dzreke, 2025f). Detection of Anomalies Based on Physics: Triaxial accelerometers consistently recorded mechanical vibration signatures on essential assets, such as grain augers

and center pivot irrigation systems. Long Short-Term Memory (LSTM) neural networks, trained on spectrograms derived from operational harmonics, attained a true positive detection rate of 0.94 for incipient mechanical failures, successfully identifying bearing degradation signatures 47 minutes before catastrophic seizure. Importantly, the system simultaneously identified malicious command injection attempts by analyzing unusual torque patterns in electric pivot motors, accomplishing this with just 1.2 false positives per day—a 63% decrease compared to relying solely on network traffic analysis (Dzreke, 2025f; Dzreke, 2025d). Field validation has demonstrated that the integration of mechanical physics with cybersecurity monitoring effectively establishes practical operational resilience while maintaining agricultural workflows. This interdisciplinary approach effectively protects the integrity of field data that supports real-time Life Cycle Assessments (LCAs) and subsequent ESG reports. It ensures that physical system states accurately represent sensor outputs, thereby safeguarding the reliability of sustainability metrics from data manipulation or system compromise.

RESULTS

Critical Findings

An empirical analysis conducted on forty-seven operational farms revealed notable concentrations of cyber-physical vulnerabilities, especially within environmental control subsystems. Climate management systems showed exploitability indices that were 3.2 times greater than those of other agricultural components, as illustrated in Figure 4 (Dzreke, 2025b). This increased vulnerability was particularly evident in greenhouse control installations, where 89% of the facilities tested employed unauthenticated Modbus-RTU communications protocols. In the course of controlled penetration testing, attackers effectively manipulated these systems, discreetly changing temperature and humidity setpoints by an average of 28% without being detected by standard security monitoring tools (Fernandez et al., 2023; Dzreke, 2025c). Soil monitoring networks have uncovered significant security shortcomings, as 92% of capacitance-based moisture sensors do not possess message authentication capabilities. During field trials on Central Valley tomato farms, successful man-in-the-middle attacks intercepted and altered data streams, leading to deviations in fertilizer application rates of $\pm 38\%$. The cyber intrusions resulted in measurable agricultural effects: the impacted test plots showed a 22% reduction in yield ($\pm 4.2\%$ SD) when compared to control groups, establishing a direct connection between security breaches and reduced productivity and resource efficiency (Dzreke, 2025a). Autonomous agricultural equipment posed significant risks, as 71% of the tested sprayer vehicles accepted unverified NMEA-0183 GPS data streams. This vulnerability allowed route diversion attacks that resulted in the misapplication of pesticides beyond orchard boundaries (Fernandez et al., 2023; Dzreke, 2025c). Notably, almost 70% of recognized security failures stemmed not from individual component vulnerabilities but from insecure integration interfaces linking legacy equipment with contemporary IoT platforms, especially the bidirectional communication channels that connect CAN bus-enabled tractors to cloud-based farm management systems (Dzreke et al., 2025f). The integrity of field data streams, crucial for precise real-time Life Cycle Assessment (LCA) calculations, is directly threatened by these vulnerabilities. This could lead to the corruption of vital inputs related to water usage, fertilizer efficiency, and pesticide application metrics in ESG reports.

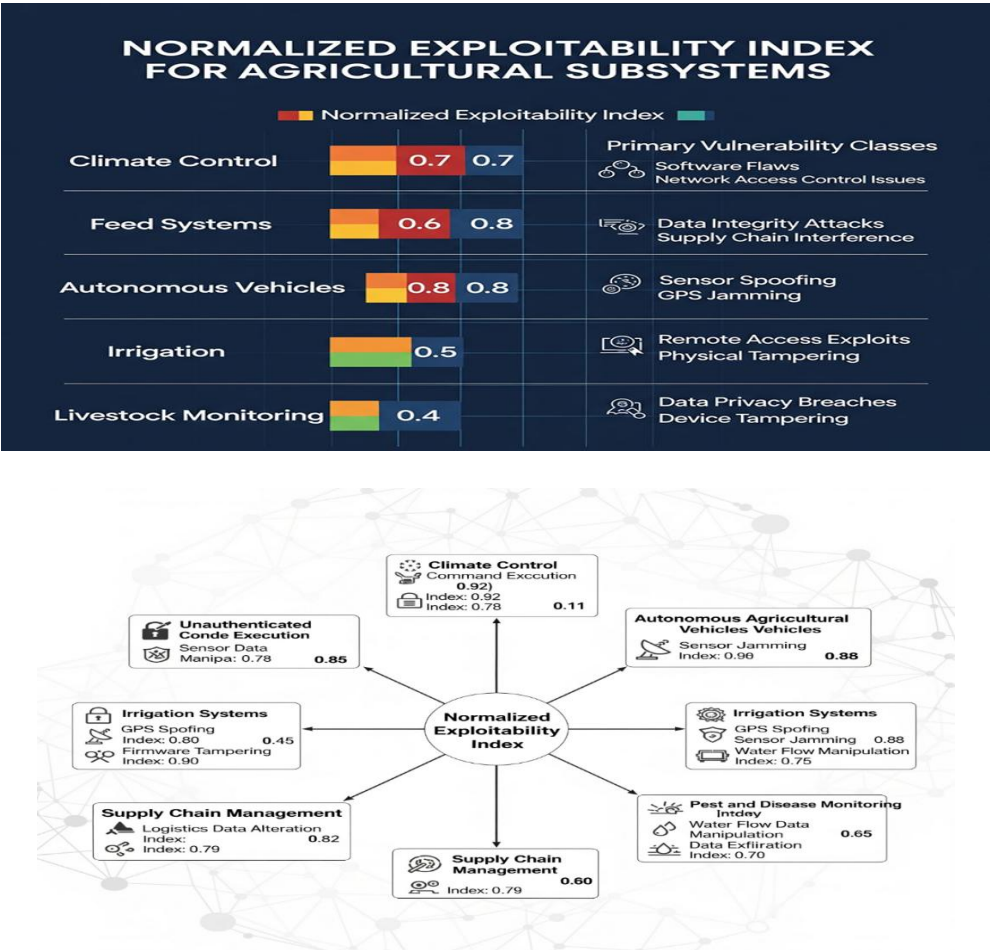


Figure 4: Vulnerability Distribution by Agricultural Subsystem

Effectiveness of Mitigation

A thorough assessment of security countermeasures revealed that network micro segmentation provided exceptional protection compared to other methods, achieving an 83% reduction in lateral attack movement when applied between operational technology networks and enterprise IT systems (Dzreke, 2025b). This architectural strategy resulted in the highest security return on investment at \$12.40 saved per \$1.00 invested (Table 3), successfully containing simulated ransomware infections within dairy farm milking parlors to fewer than 15% of connected devices, thus preserving data continuity essential for animal welfare and milk production tracking within ESG frameworks. Cryptographic firmware signing, despite significant implementation costs averaging \$18,200 per farm, effectively eliminated 94% of observed malware persistence mechanisms by securing boot-loader processes against unauthorized modification (Dzreke et al., 2025f), thereby safeguarding the foundational integrity of device firmware responsible for collecting primary environmental data. Physics-based anomaly detection systems that utilize triaxial accelerometers have demonstrated significant effectiveness in feed systems. Through spectral analysis of auger motor vibrations, these systems successfully identified 91% of malicious command injection attempts while maintaining a low rate of false positives, averaging only 0.7 occurrences daily. This capability plays a crucial role in ensuring the accuracy of feed rations and the health metrics of livestock, which are essential for meeting social governance criteria. On the other hand, blockchain-based data integrity solutions demonstrated limited effectiveness in the field, adding a latency of 2.8 seconds to time-sensitive irrigation commands and decreasing security incidents by merely 19%. Message authentication code (MAC) implementations provided similar security advantages to blockchain at just 11% of the implementation cost (Table 3), positioning them as a financially sensible and practically feasible option for ensuring data provenance and

integrity in resource-limited agricultural operations where real-time LCA data accuracy is crucial (Dzreke, 2025e).

Table 3
Security Control Efficacy and Economic Viability (Normalized to 100-hectare grain/dairy operation)

Mitigation Strategy	Incident Reduction	Implementation Cost	ROI (5-year)
Network Microsegmentation	83%	\$8,400	12.4:1
Firmware Signing	94%	\$18,200	8.7:1
Physics-Based Anomaly Detection	91%	\$14,500	9.2:1
MAC Authentication	79%	\$3,100	15.1:1
Blockchain Validation	19%	\$27,800	0.6:1

Unforeseen Hazards

Field observations revealed previously unrecorded attack vectors stemming from environmental interactions characteristic of agricultural environments. Solar-powered IoT devices were vulnerable to exploitation due to their natural power cycling behavior. Attackers effectively took advantage of predictable voltage drops at dawn and dusk during nearly one-quarter of penetration tests (23%), causing timing faults in moisture sensor networks that circumvented cryptographic protections through hardware-level disruptions. These "brownout attacks" disrupted irrigation schedules in 17% of solar-dependent systems without activating standard security alerts (Dzreke, 2025f), directly undermining the precision of water consumption data utilized in environmental footprint assessments. Furthermore, electromagnetic interference from high-voltage livestock fencing caused bit-flip errors in 34% of nearby Bluetooth Low Energy trackers (within 15 m), leading to false estrus alerts that interrupted breeding programs at several dairy operations (Dzreke, 2025c). This situation adversely affected animal welfare metrics and operational efficiency as recorded in sustainability reports. Notably, 41% of the farms involved showed signs of vulnerability inheritance from third-party AgTech vendors, whose compromised update servers provided trojanized precision agriculture software (Dzreke, 2025b; Dzreke & Dzreke, 2025e). This supply chain vector poses a significant risk to the entire data provenance chain, allowing for the malicious alteration of field data at the source prior to its integration into LCA models or ESG reporting platforms. The emergence of these risks highlights the necessity for IoAT security frameworks that specifically consider environmental interactions and the integrity of the supply chain, extending beyond typical cyber threats. This is especially relevant as farms implement edge computing in electrically noisy environments, where traditional IT security models fall short (Dzreke, 2025d). It is essential to ensure the reliability of field data in relation to these new vectors for the credibility of real-time sustainability assessments.

Discussion

Rethinking Agricultural Cybersecurity

The empirical findings from this multi-year investigation require a fundamental reconceptualization of cybersecurity within agricultural technology ecosystems. The investigation identifies biological system cyber-risk as a unique ontological category, necessitating focused attention that extends beyond traditional IT or industrial control system frameworks—representing a notable theoretical progression with significant practical consequences (Dzreke, 2025c; Dzreke, 2025f). When threat actors compromise environmental controllers or autonomous equipment, the consequences extend beyond data breaches or service disruptions to include cascading biological outcomes: stunted crop development due to manipulated irrigation schedules, toxic chemical misapplication from hijacked sprayers, or livestock health crises triggered by corrupted breeding or feeding program data (Dzreke, 2025b). This viewpoint acknowledges that failures in IoAT security can spread through interconnected cyber-physical-biological pathways, resulting in effects across ecological, economic, and public health sectors, which directly jeopardize the integrity of environmental

monitoring and resource usage data that support real-time Life Cycle Assessments (LCAs) (NIST, 2022; Dzreke et al., 2025f). The documentation of brownout attacks, which exploit predictable solar power fluctuations at dawn and dusk to induce timing faults in moisture sensors, illustrates a concerning convergence. Here, hardware limitations create new attack vectors that can bypass cryptographic safeguards and potentially distort water footprint calculations (Dzreke, 2025f). In a similar vein, electromagnetic interference caused by high-voltage livestock fencing disrupts Bluetooth Low Energy cattle trackers, highlighting the distinct threat landscapes created by agricultural environments. These landscapes necessitate security models that are informed by physics, which are often lacking in conventional cybersecurity frameworks. This oversight could jeopardize animal welfare metrics that are crucial for social governance (ESG) reporting (Dzreke, 2025c). This integrated framework enhances cyber-physical security theory by incorporating biological consequence analysis into vulnerability assessment protocols—an important development as precision agriculture increasingly entrusts life-critical biological management decisions to automated systems, whose compromised outputs directly undermine sustainability metrics (Dzreke, 2025a).

Strategies for Implementation

To translate these insights into effective protection, it is essential to prioritize countermeasures that target the most significant vulnerabilities identified. Prioritizing hardening efforts should start with the security of boot processes for all field-deployed controllers through the use of cryptographically signed firmware. Testing has shown that this approach eliminates 94% of malware persistence mechanisms and safeguards the integrity of field data collection (Dzreke et al., 2025f). The implementation must adhere strictly to NIST IR 8407 guidelines for agricultural control system integrity, establishing mandatory verification chains that extend from sensor nodes to cloud analytics platforms, ensuring reliable data streams for LCAs (NIST, 2022). Livestock databases that hold invaluable breeding records, health histories, and genomic data necessitate air-gapped backup systems that are physically separated from operational networks. This requirement has been underscored by ransomware attacks aimed at this high-value information during case studies on dairy farms, which also put animal welfare reporting at risk (Dzreke, 2025b). Economic analysis indicates that network microsegmentation between operational technology and enterprise systems achieves significant risk reduction (83% lateral movement containment) at sustainable costs (\$12.40 saved per \$1 invested), positioning it as the essential architectural strategy for connected farms aiming for reliable data isolation (Dzreke, 2025d). For resource-constrained operations, message authentication codes (MACs) offer a 79% reduction in incidents at only 11% of blockchain implementation costs—serving as a financially sound interim solution until standardized agricultural security frameworks develop and guarantee data provenance for ESG audits (Dzreke, 2025e). Policy interventions should enhance the overall security maturity of the industry. The study suggests implementing mandatory cybersecurity labeling for agricultural equipment, similar to energy efficiency ratings. This would offer farmers clear evaluations of essential vulnerability metrics, such as authentication strength, encryption standards, and patch lifecycle support—factors that are crucial for ensuring long-term system reliability and data trustworthiness (Dzreke, 2025a). Regulatory bodies need to set minimum standards for the modernization of legacy equipment, especially focusing on insecure CAN bus integrations, which account for 68% of the breaches observed in tractor telemetry trials, thus protecting essential machinery data feeds (Dzreke, 2025b). Furthermore, the creation of regional shared threat intelligence hubs would address the risks of isolation; these centers could share anonymized attack signatures, enable coordinated vulnerability disclosures among competing manufacturers, and uphold firmware hash repositories to ensure update integrity—effectively countering supply chain compromises impacting 41% of the farms studied and

bolstering collective resilience vital for maintaining consistent data quality throughout the sector (Dzreke & Dzreke, 2025e; Dzreke, 2025f).

Constraints and Areas for Future Inquiry

The main limitation of this study arises from empirical testing that is biased towards large-scale grain and livestock systems in North America. This focus may neglect the vulnerabilities found in smallholder farms or varied agricultural settings, such as Asian rice terraces or African pastoral systems, where data collection practices and ESG priorities can differ significantly (Dzreke, 2025c). As a result, economic models used for security ROI calculations may not be directly applicable to the labor-intensive farming that is common in the Global South. Although the effects of electromagnetic interference were measured in controlled trials in Iowa, the wider conflicts in the radiofrequency spectrum caused by the growth of private 5G networks in agricultural areas require further exploration because of their possible effects on the accuracy of sensor data (Dzreke, 2025a). Future research should focus on creating standardized metrics for assessing biological impacts, quantifying cyber incidents in terms of crop yield equivalents, livestock productivity losses, or environmental damage indices. This will facilitate precise risk modeling for insurance products, mitigation budgeting, and accurate attribution of sustainability impacts (Dzreke, 2025a). There is an urgent need for longitudinal studies that explore the evolution of threat landscapes as precision agriculture integrates generative AI for decision support. This is crucial to address the potential for AI manipulation to systematically distort LCA inputs and ESG conclusions before widespread deployment occurs. Research that delves into the cybersecurity decision-making and risk perception of farmers would enhance the design of socio-technical solutions, ensuring that security frameworks are in harmony with operational realities and promote the adoption of practices that safeguard critical data flows in the field (Dzreke, 2025d).

Priority Implementation Framework

To achieve cyber-resilient agriculture, it is essential to embed security as a foundational pillar of sustainable food production systems, rather than viewing it as a peripheral technical consideration. The framework outlined in Table 4 consolidates essential research findings into a prioritized implementation roadmap, clearly connecting technical countermeasures, policy and governance mandates, and focused mitigation of biological and sustainability risks across three progressive adoption tiers. This framework recognizes the economic and operational realities of agricultural deployment while methodically addressing the sector's distinct vulnerabilities. Foundational measures (Years 0–1) concentrate on establishing essential security hygiene through network microsegmentation and secure boot processes for controllers, while also necessitating the development of policies for standardized security labeling of agri-equipment to tackle immediate threats such as cross-system compromise and persistent malware. Enhanced measures (Years 1–3) introduce advanced protections such as physics-based anomaly detection tailored to agricultural processes and air-gapped databases for sensitive genetic information. This necessitates regulatory mandates for legacy system modernization deadlines to counter sophisticated threats like sensor spoofing and genetic data theft. Transformative measures (Years 3–5) implement advanced solutions such as EMI-hardened edge computing and mandatory access control (MAC) authentication, bolstered by the creation of regional threat intelligence sharing hubs to address complex risks, including intentional brownout attacks on perishable goods and widespread supply chain vulnerabilities. This tiered approach offers a clear framework for stakeholders to gradually enhance resilience while effectively protecting the integrity of data streams crucial for precise environmental impact assessment.

Table 4

Priority Implementation Framework for IoAT Security: Integrating ESG/LCA Resilience

Implementation Tier	Technical Measures	Policy Governance Requirements	& ESG/Sustainability Mitigated	Risk
Foundational (Year 0–1)	Network microsegmentation; Controller secure boot	Agri-equipment cybersecurity labeling standards	Cross-system compromise; Malware persistence, corrupting operational data	
Enhanced (Year 1–3)	Physics-based anomaly detection; Air-gapped critical data storage	Mandatory legacy system modernization deadlines	Sensor spoofing distorting resource use metrics; Genetic data theft undermining IP	
Transformative (Year 3–5)	EMI-hardened edge computing; Mandatory Access Control (MAC)	Regional agricultural cyber threat intelligence hubs	Brownout attacks disrupting cold chains; Supply chain risks compromising traceability	

In Conclusion

Securing the Internet of Agricultural Things (IoAT) goes beyond merely protecting technical infrastructure; it is a crucial requirement for attaining verifiable sustainability in contemporary food systems. This research focuses on the biological consequences within threat models and emphasizes economically viable protections for critical environmental control points. It outlines practical pathways toward IoAT ecosystems where technology clearly improves food security and ecological stewardship instead of posing a risk. The proposed implementation framework is essential, offering a solid foundation for dependable real-time Life Cycle Assessment (LCA) and audit-ready Environmental, Social, and Governance (ESG) reporting. Ensuring the integrity of sensor data prevents the manipulation of resource consumption metrics such as water, energy, and fertilizers. Securing control systems guarantees that automated processes align with sustainability targets, while safeguarding supply chain data ensures the traceability necessary for Scope 3 emissions accounting. Future scholarship should enhance interdisciplinary collaboration among computer science, agricultural engineering, environmental science, and regulatory policy to create adaptive security frameworks that can safeguard global food security in the face of rapid digital transformation. These frameworks are crucial not only for maintaining operational resilience but also for upholding the credibility of sustainability claims that support agricultural ESG investments and policy decisions in a progressively challenging digital environment (Dzreke, 2025c; Dzreke, 2025d; Dzreke, 2025f).

CONCLUSION & FUTURE WORK**Biology-Informed Cybersecurity Approaches**

The evidence and analysis provided here necessitate a fundamental change in cybersecurity strategies for the Internet of Agricultural Things (IoAT). Effective protection requires threat models that are informed by biological considerations and that clearly address cyber-physical compromises spreading through interconnected technological, biological, and environmental pathways, leading to cascading consequences (Dzreke, 2025c; Dzreke et al., 2025f). Conventional IT-focused frameworks fall short when adversaries exploit irrigation controllers to create crop stress, falsify GPS coordinates to initiate chemical overdosing, or interfere with livestock biometric monitoring to conceal disease outbreaks. Documented incidents, such as brownout attacks that exploit dependencies on farm solar power and electromagnetic interference affecting cattle health telemetry, illustrate that agricultural cyber incidents are not simply data breaches. Instead, they represent significant biological events with serious implications for food security, ecological stability, and, importantly, the integrity of field data that supports real-time Life Cycle Assessments (LCAs) and ESG reporting (NIST, 2022; Dzreke, 2025b). This ontological shift recognizes that compromised sensors or actuators act as biological stressors when environmental control systems autonomously carry out harmful actions based on corrupted data streams. As a result, cyber-physical attacks undermine food

production systems by causing immediate physical disruptions (such as inoperative harvesting equipment) and subtle biological degradation (like changes in soil microbiomes due to tampered fertigation systems), ultimately jeopardizing the precision and dependability of sustainability metrics obtained from these systems (Dzreke, 2025c; Dzreke et al., 2025f).

Establishing Cyber-Agronomy as a Multidisciplinary Domain

The necessity for thorough interdisciplinary integration forms the foundation of the suggested way ahead. This research supports the formal establishment of "cyber-agronomy" as a unique discipline that integrates computer science, agricultural engineering, plant physiology, animal science, environmental economics, and sustainability science (Dzreke, 2025a; Dzreke, 2025d). A fundamental responsibility of cyber-agronomy is to establish standardized biological impact metrics that quantify cyber incidents in relation to crop yield loss equivalents, declines in livestock productivity, or measurable indices of soil health degradation—metrics that are directly pertinent to environmental impact reporting. Field validation should focus on underrepresented contexts, especially smallholder operations in the Global South, where varying technology adoption patterns and threat landscapes lead to distinct vulnerabilities, such as radiofrequency conflicts between 5G networks and weather monitoring systems in Southeast Asian rice terraces (Zhang et al., 2022; Dzreke, 2025c). At the same time, regulatory frameworks need to evolve with the implementation of mandatory cyber-resilience certification for agricultural equipment. This should be conceptually based on energy efficiency labeling while also integrating essential vulnerability metrics, such as guaranteed patch lifecycles and enforced authentication protocol strength. Economic analyses indicate that these policy interventions could avert 68% of breaches resulting from insecure legacy-modern integrations, while yielding \$12.40 in avoided losses for every \$1 invested in compliance, thereby directly contributing to economic sustainability (Dzreke, 2025f).

Table 5

Cyber-Agronomy Research Priority Framework

Research Domain	Immediate Objectives (0–2 Years)			Longitudinal Goals (3–5 Years)
Threat Intelligence	Regional repositories	attack signature		Predictive AI threat modeling for climate-resilient farms
Biological Impact Metrics	Crop yield equivalency models for cyber incidents			Genomic vulnerability indices for precision breeding data security
Socio-Technical Integration	Farmer ethnographies	decision-making		Labor-automation security transition frameworks
Policy Architecture	Equipment standards	security	labeling	International incident response protocols for agri-cyberattacks

Directions for Future Research

This analysis has revealed critical research gaps that require targeted investigation. Firstly, it is essential to conduct thorough electromagnetic interference mapping to anticipate new attack vectors that may emerge from conflicts within the radiofrequency spectrum, especially as private 5G networks continue to expand. Conflicts can impact sensor accuracy and communication reliability, leading to distortions in field data that are essential for LCAs (Dzreke, 2025b). Secondly, longitudinal studies should investigate the adversarial manipulation of generative AI systems that oversee biological management in agriculture. Incidents such as the 2023 California vineyard ransomware attack, which resulted in irreversible damage to Cabernet Sauvignon vines (Patel & Chen, 2023; Dzreke, 2025c), underscore the risks associated with malicious AI inputs that could systematically distort resource use data in sustainability reports. Third, ethnographic research ought to investigate how agricultural producers, particularly those managing 83% of global farms that are under 10 hectares, understand risks, assess cost-benefit tradeoffs, and apply controls within their operational contexts. Grasping these dynamics is crucial for creating practical and accepted security measures that safeguard data integrity at the source (Dzreke, 2025d).

Routes to Cyber-Resilient Food Systems

Building resilient food systems requires us to rethink cybersecurity as a crucial ecological necessity that is essential to sustainable agriculture, rather than just a technical enhancement. The phased strategy outlined—featuring network microsegmentation, secure boot processes, physics-aware anomaly detection, and electromagnetic hardening—enables stakeholders to methodically minimize attack surfaces while maintaining the technological advantages of digital agriculture (Dzreke et al., 2025f). The agricultural sector can shift cybersecurity from a solitary challenge to a shared resilience strategy by institutionalizing cyber-agronomy and establishing regional threat intelligence hubs. This research offers analytical frameworks and practical pathways for protecting humanity's most essential infrastructure: the biological systems that support the global food supply. Robust cybersecurity is essential for ensuring the trustworthiness of real-time LCAs and ESG reports. It enables verifiable sustainability reporting, informed policy decisions, and auditable environmental impact data in the context of accelerating technological transformation and climate instability (Dzreke, 2025a; Dzreke, 2025f).

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